Falsification-as-code: hash-frozen success criteria, and nine ways a pre-registered criterion still fails

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Abstract

Pre-registration is normally a document. We report a working protocol in which it is a *hash*: the block of success criteria lives in the analysis script's own docstring, is frozen by SHA-256 before execution, and a verifier fails the build if it has moved afterwards. Amendments are possible but public. Over one cosmology corpus — 88 frozen criteria, 198 numbered register entries, 78 adjudicated attacks — the protocol behaved very differently from what its designers expected. It did not mainly catch *wrong results*; it caught *ill-posed criteria*. We give a taxonomy of nine distinct ways a pre-registered criterion can be formally satisfied while establishing nothing, each with a worked example that occurred: a criterion on a non-identifiable parameter; a bound unfit for the data's binning; a comparison against an unoptimised reference; the wrong comparator; a boundary minimum on a branch that was closed; two mutually incompatible requirements frozen together; a docstring asserting 60 000 grid points while its own code used 240 001; a validation passing at 10^{-4} relative while measuring a quantity that does not exist; and a verdict decided by the fourteenth decimal of a floating-point comparison. We then report the countermeasures that survived contact: mutation-testing the guardrails themselves, generic rather than hand-written detection of unsubstituted markers, regular expressions instead of string equality, testing the refuting branch before the confirming one, and declaring the direction of every substitution in advance. The protocol's cost is that studies stop themselves; over the corpus, five did. Its benefit is that 69 claims were retracted or downgraded, several of them the corpus's own headline results, and most of them by the analyst's own machinery rather than by a referee. We claim no generality beyond $N = 1$, and we report what the protocol does *not* catch.

1 The protocol

The unit is a script. Its docstring opens with a block headed CRITÈRES PRÉ-ENREGISTRÉS, containing: the validations that must pass before anything is published, the criteria themselves — required to be *exhaustive* and *exclusive*, so that every possible outcome maps to exactly one written verdict — and a declaration of what the study will *not* claim.

Three commands operate on it:

- `freeze` extracts the docstring by AST, hashes it, and records `path` $\rightarrow$ `sha256` in a lock file;
- `verify` recomputes and exits non-zero if any hash moved — blocking in CI;
- `freeze --amend` allows a criterion to change, but writes the old and new hashes, with a mandatory justification, to a public retraction file.

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Two consequences follow immediately and are worth stating because they are the whole point. First, editing the criteria *after* seeing the result is not forbidden — it is *recorded.* Over the corpus reported here, the amendment channel was used zero times: 88 frozen criteria all carry the hash of their first freeze. Second, the body of a script may be corrected freely; only the criteria are immutable. This separation is what makes the protocol usable, and it is also where the first surprise came from.

2 The surprise: it caught criteria, not results

We expected the protocol to catch results that drifted from their stated success condition. It did that rarely. What it caught, repeatedly, was criteria that were *formally satisfiable without establishing anything.* Nine distinct failure modes occurred, and they do not reduce to carelessness — each was written by an analyst trying to be rigorous, and each passed a reading at the time it was frozen.

Two further modes appeared during the writing of this paper and belong in the taxonomy although we have not numbered them:

- **a frozen definition diverging from the operative one.** A verification tool was frozen with a declared sky-region definition differing from the one the corpus used everywhere else by $10°$ in right ascension. It counted 412 objects where the corpus counted 416. The tool was right about the defect it had found and wrong about the definition it used to find it.
- **a validation blocked by its own resolution.** A test required an uncertainty to shrink by a factor k; the scan grid could not resolve an uncertainty smaller than one grid step, so at $k = 4$ the measured factor was 2.0 instead of 2.8–5.6. The validation correctly refused to conclude on a measurement its own mesh could not deliver. Notably, an *independent* signature in the same run — the χ^2 gains scaling as k^2 to three decimals — did establish that the manipulation was operative. We did not substitute it: doing so is failure mode 7 in a different costume.

3 What actually works

The countermeasures below are reported because they survived; several plausible-sounding alternatives did not.

Mutation-test the guardrails. A verification tool returning "55/55, no blocking failure" establishes nothing until it is shown capable of failing. We injected five faults into the data files the tool reads — a gain inflated by 5%, a weighted mean by 2%, a central value displaced, an uncertainty multiplied by four, a fraction pushed out of bounds — and required all five to be caught. All five were. Only then did the 55/55 acquire meaning. *We recommend this as mandatory, not optional.* It is the direct countermeasure to failure mode 8.

Detect generically, never from a hand-written list. A template-substitution check enumerated five placeholder names by hand. A sixth was added to the template later; the check passed; the published page failed at line 1 with every placeholder unsubstituted. The replacement check is a single regular expression over `__[A-Z_]+__` and cannot miss a placeholder it has never heard of. The general rule: *a check must be able to detect the case its author did not think of.*

Regular expressions, never string equality, on text. A guardrail written as an equality test against a line of prose broke silently when the file gained a line break. Every textual check in the corpus is now a regex insensitive to line breaks. This is rule 7 of the corpus's own constitution and it was written after the failure, not before.

Test the refuting branch first. Where a study is designed to confirm one of the analyst's own readings, the code evaluates the branch that would refute it *before* the branch that would confirm it, and the confirmation threshold is set stricter than the refutation threshold (in our case a factor 3 against a factor 1). Twice this ordering mattered: a study of ours returned EXPLICATION RÉFUTÉE on a mechanism we had asserted in two previous entries without ever testing it.

Declare the direction of every substitution. Where a value must be assumed, the protocol requires it to be chosen *against* the thesis being defended, and requires the choice to say so. A conversion factor of 0.8389 was retained over an equally defensible 0.8429 on exactly this ground; stating the reason is what distinguishes a conservative choice from a lucky one.

Let the study stop itself, and publish the stop. Five studies in this corpus halted on their own validations and published nothing. In each case the stop was informative: one revealed that an anchor was being compared at the wrong optimum, another that a grid could not resolve what it was asked to measure. *A protocol whose studies never stop is not being enforced.*

4 Cost and benefit

The cost is real. Studies stop. Criteria that were reasonable when written turn out to be ill-posed and cannot be repaired in place — the corpus's rule is that a frozen criterion is a *record*, not a claim to be rewritten, so a superseded tool is replaced rather than amended, and both remain in the register. This produces visible clutter, and we consider the clutter a feature: the amendment count stayed at zero not because criteria were always right but because being wrong was survivable without editing them.

The benefit is measured on one axis we can report honestly: 69 claims were refuted, downgraded or corrected, and the majority were caught by the analyst's own machinery. Several were the corpus's headline results:

- a model leading a 25-entry comparison table was found to owe 8.62 of its 9.84 units of advantage to a calibration artefact — after three adversarial audits had validated it without leaving the flawed pipeline;
- a sky-area fraction was found to be wrong by a factor $\pi/2$, having been copied through six generations of artefacts, with the effect that the corpus had been *overstating its own anomaly* by 57%;
- a mechanism the analyst had asserted in two successive entries was refuted by the first test ever applied to it.

We note without irony that the third of these is the protocol working exactly as intended and the analyst not having done so.

5 What the protocol does not catch

It is mechanical. It does not read equations, does not judge physics, and cannot tell whether a model is interesting. In this corpus, the most consequential single audit was triggered not by any tool but by a human reader saying the results looked *too good to be true*; that audit found a boundary minimum (failure mode 5) that every automated check had passed. A later literature verification killed a planned paper outright by establishing that all four of its claimed results had been published, the oldest in 2015 and the newest four days earlier. Neither of those is reachable by hashing.

There is also a specific hazard the protocol creates. Because criteria are immutable, there is pressure to write them loosely enough to be safe. We countered this by requiring criteria to be *exhaustive and exclusive* — every outcome named in advance, including the ones that refute the analyst — but we cannot claim the pressure is eliminated, only that it is visible.

6 Reproducibility

The protocol is 120 lines of Python and depends on nothing but the standard library. Its own success criteria are frozen by itself (commit 0): it succeeds if freeze-then-verify passes on at least two real scripts and if falsifying a criterion is detected with a non-zero exit; it fails and is not published if a false positive appears on an unmodified file. Both conditions have held over 88 frozen criteria.

#	failure mode	what happened
1	non-identifiable parameter	A criterion was frozen on a quantity the data could not constrain. It was satisfiable by any outcome, so it excluded nothing.
2	bound unfit for the binning	A goodness-of-fit bound of the form $\chi^2/(N+17)$ was frozen for a dataset with 22 bins. The offset was inherited from a different data configuration and made the bound meaningless at that N.
3	unoptimised reference	A validation compared the model at a *fixed* parameter point against a reference at its *optimum*. The gap it measured was the optimisation, not the physics.
4	wrong comparator	A validation was run against a comparator that reintroduced a radiation term the model had removed. It failed for a reason unrelated to the hypothesis.
5	boundary minimum	A criterion returned a very tight constraint. Re-examination showed the minimum sat on the edge of the accessible domain: one branch of the parameter was closed by the implementation, so the constraint was one-sided while being reported as two-sided.
6	mutually incompatible requirements	Two constraints were frozen in the same block — a grid of 41 points, and a tolerance of ± 0.3 — that could not both be met. The study could not succeed by construction.
7	docstring/code divergence	A docstring asserted "a dense grid (60 000 points, log)"; the code, after successive corrections to the body, used 240 001. **The criterion would have been satisfied by something other than what it stated.**
8	a check that measures nothing	A validation required a conversion factor to be constant to better than 2%. It passed at 0.010%. The quantity it converted did not exist: the source parameter had been misread, and there was no lever arm to convert. *A validation that passes perfectly while measuring a phantom is more dangerous than one that fails.*
9	a verdict decided by rounding	Two thresholds were compared with `>=` against floating-point grid arithmetic. The measured values were exactly 2 and exactly 1; the computed values were 1.9999999999999791 and 0.9999999999999836. Both comparisons failed, and both failures fell in the direction convenient to the analyst.

Table 1: Nine ways a pre-registered criterion was formally satisfied, or formally decided, without establishing what it was written to establish. Every row occurred in one corpus over two days of intensive use.